

Document Details	
Document No.	830016
Document Rev.	01
Document Title	ACM700X-X Programming and Test Instructions

Revision History			
Rev	Description	Date	By
01	Initial Revision (Bin Version 4.13.5)	4/24/25	KB

Support Equipment Required

Name	Specification	Part Number	Qty Required
USB-A to RS-232 Adapter	Any Length	440064	1
DB9 to RJ45 Adapter	Cisco Straight	319015	1
RJ45 Cable	Any Length	810042	1
RJ45 Loopback Cable	Max 6" Length	800060	2 (SKU Dependent)
4-Pin Power Supply	12V 1.25A Output	450039	1
Serial Loopback	Black Cisco Straight	449055	7 (SKU Dependent)
USB-A Drives	Max 2" Length	569044	5

SETUP INSTRUCTIONS:

1. Insert DB9 to RJ45 Adapter to the USB-A to RS-232 Adapter DB9 female port (**Figure 1**)
2. Insert RJ45 Cable to the DB9 to RJ45 Adapter (**Figure 1**)

OPERATOR INSTRUCTIONS:

1. Insert the RJ45 cable from the DB9 to RJ45 Adapter to the top-left RJ45-serial port 1 on the PCA required for testing (**Figure 1**)
2. Insert 4-Pin Power Supply to the external 4-pin female port (**Figure 2**)
3. Insert the black Serial Loopback to the remaining RJ45-serial ports (**Figure 2**)
4. Insert USB-A Drives to the USB-A ports except the bottom internal USB port (**Figure 2**)
5. Insert RJ45 Loopback Cable(s) to all ethernet ports by looping each other (**Figure 3**) (**Figure 4**)
6. Wait for “acm700X-X login:” to appear in the terminal (**Figure 5**)

Note: This takes approximately 5-7 minutes

7. Log into the PCA using the following credentials until the “#” prompt is seen (**Figure 5**)
acm700X-X login: **root**
Password: **default**
8. Check USB Drives for five “/dev/sdx” storage devices by using the following command (**Figure 6**)
ls /dev/sd[a-e]
9. Check (Serial Port 2-4) 3 serial ports return "LLL" characters by using the following command (**Figure 7**)
timeout 2s loopback /dev/port0[2-4] -q
10. If 8-Serial PCA (Serial Port 5-8), Check the remaining 4 serial ports returned "LLLL" characters (**Figure 8**)
timeout 2s loopback /dev/port0[5-8] -q
11. If 2-Ethernet PCA, Check the speed returned “Speed: 1000M” (**Figure 9**)
ifconfig eth1 up&&sleep 5&ðtool eth1
12. If 5-Ethernet, Check each port returned “1000Mb/s” (**Figure 10**)
/etc/scripts/swtool diag
13. Ensure two MAC addresses have been assigned in the matrix provided by Opengear
14. Print the first MAC address on an adhesive label
15. Append & scan MAC 1 adhesive label and append MAC 2 after each command (**Figure 11**)
setfset -u ethaddr=<“MAC 1 ADDRESS”>
setfset -u eth1addr=<“MAC 2 ADDRESS”>
Example Below (do not execute commands below)
setfset -u ethaddr=00:13:C6:13:19:0C
setfset -u eth1addr=00:13:C6:13:19:0D
16. Apply MAC 1 address adhesive label on top the of serial ports or near serial ports (**Figure 12**)
17. Remove all accessories and repeat operator instructions for additional PCAs

FIGURE 1

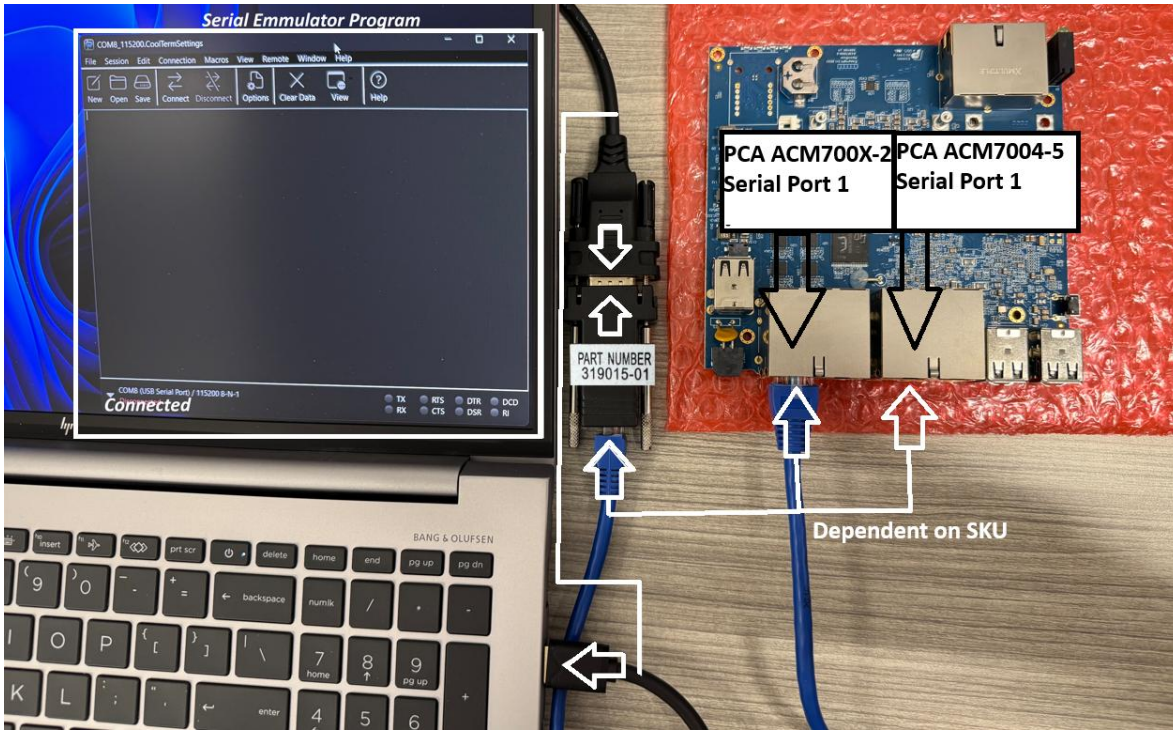


FIGURE 2

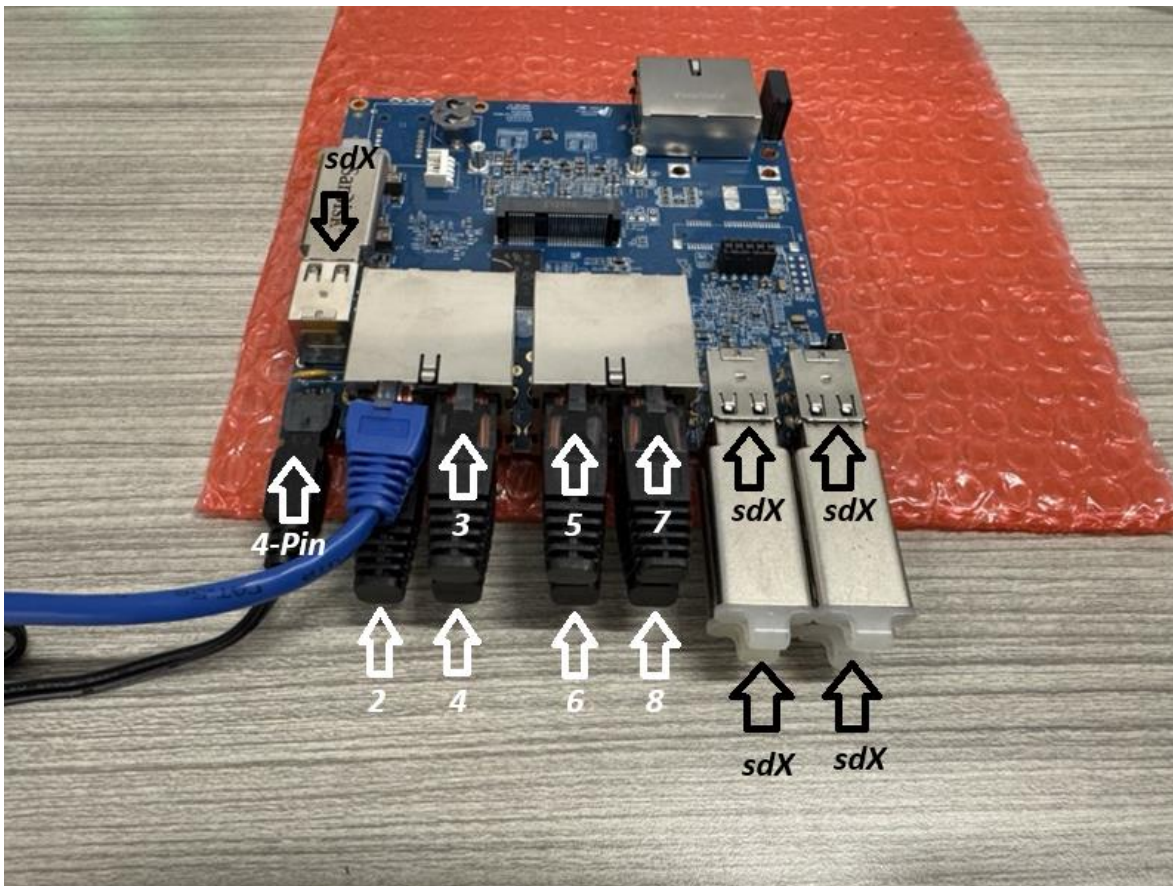


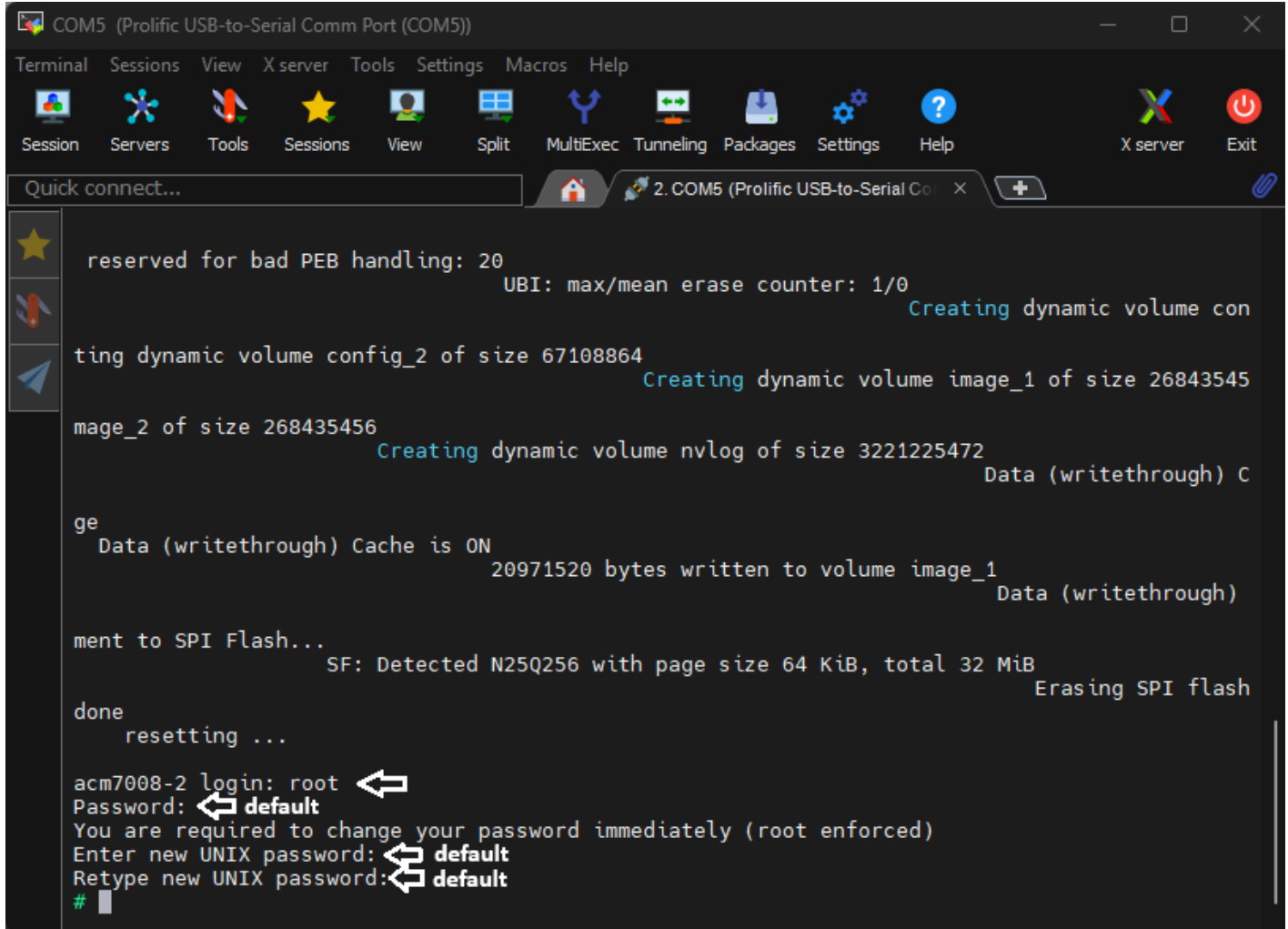
FIGURE 3



FIGURE 4

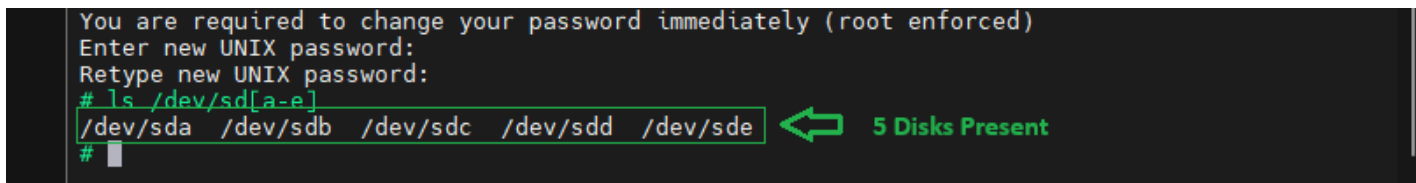


FIGURE 5



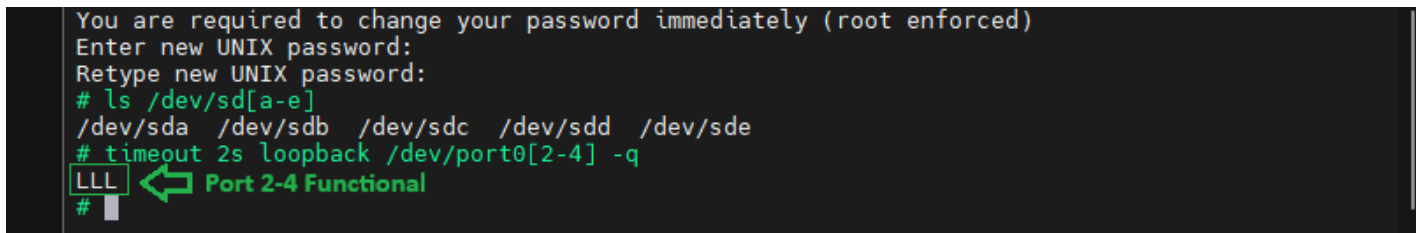
```
COM5 (Prolific USB-to-Serial Comm Port (COM5))
Terminal Sessions View X server Tools Settings Macros Help
Session Servers Tools Sessions View Split MultiExec Tunneling Packages Settings Help X server Exit
Quick connect... 2. COM5 (Prolific USB-to-Serial Co x
reserved for bad PEB handling: 20
UBI: max/mean erase counter: 1/0
Creating dynamic volume con
ting dynamic volume config_2 of size 67108864
Creating dynamic volume image_1 of size 26843545
image_2 of size 268435456
Creating dynamic volume nvlog of size 3221225472
Data (writethrough) C
ge
Data (writethrough) Cache is ON
20971520 bytes written to volume image_1
Data (writethrough)
ment to SPI Flash...
SF: Detected N25Q256 with page size 64 KiB, total 32 MiB
Erasing SPI flash
done
resetting ...
acm7008-2 login: root
Password: default
You are required to change your password immediately (root enforced)
Enter new UNIX password: default
Retype new UNIX password: default
#
```

FIGURE 6



```
You are required to change your password immediately (root enforced)
Enter new UNIX password:
Retype new UNIX password:
# ls /dev/sd[a-e]
/dev/sda /dev/sdb /dev/sdc /dev/sdd /dev/sde 5 Disks Present
#
```

FIGURE 7



```
You are required to change your password immediately (root enforced)
Enter new UNIX password:
Retype new UNIX password:
# ls /dev/sd[a-e]
/dev/sda /dev/sdb /dev/sdc /dev/sdd /dev/sde
# timeout 2s loopback /dev/port0[2-4] -q
LLL Port 2-4 Functional
#
```

FIGURE 8

```
You are required to change your password immediately (root enforced)
Enter new UNIX password:
Retype new UNIX password:
# ls /dev/sd[a-e]
/dev/sda /dev/sdb /dev/sdc /dev/sdd /dev/sde
# timeout 2s loopback /dev/port0[2-4] -q
LLL
# timeout 2s loopback /dev/port0[5-8] -q
LLLL ← Ports 5-8 Functional
#
```

FIGURE 9

```
# timeout 2s loopback /dev/port0[5-8] -q
LLLL
# ifconfig eth1 up&&sleep 5&&ethtool eth1
Settings for eth1:
    Supported ports: [ TP MII ]
    Supported link modes:   10baseT/Half 10baseT/Full
                           100baseT/Half 100baseT/Full
                           1000baseT/Half 1000baseT/Full
    Supports auto-negotiation: Yes
    Advertised link modes:  10baseT/Half 10baseT/Full
                           100baseT/Half 100baseT/Full
                           1000baseT/Half 1000baseT/Full
    Advertised auto-negotiation: Yes
    Speed: 1000Mb/s ←
    Duplex: Full
    Port: MII
    PHYAD: 1
    Transceiver: external
    Auto-negotiation: on
    Link detected: yes
#
```

FIGURE 10

```
# timeout 2s loopback /dev/port0[2-4] -q
LLL
# /etc/scripts/swtool diag
ACM7004-5 diagnostics
p1: link 1000 Mb/s mdix ←
p2: link 1000 Mb/s
p3: link 1000 Mb/s mdix
p4: link 1000 Mb/s
#
```

FIGURE 11

```
PHYAD: 1
Transceiver: external
Auto-negotiation: on
Link detected: yes
# setfset -u ethaddr=00:13:C6:13:19:0C ← Scanned From Label
# setfset -u eth1addr=00:13:C6:13:19:0D ← Copied From Matrix
#
```


FIGURE 12

